DE BRUIJN–NEWMAN HEAT-FLOW VELOCITY $\frac{\mathrm{d}\gamma_k}{\mathrm{d}\tau}\Big|_{\tau=0} \equiv \frac{1}{4}\kappa(\gamma_k)$, CROSS-TERM-FREE GAUSSIAN WIGNER FACTORIZATION, AND THE TRANSVERSE DIPOLE EVOLUTION PDE HIERARCHY FOR RIEMANN'S Ξ-FUNCTION

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Abstract. We establish three exact analytical laws connecting the De Bruijn–Newman heat-flow deformation $\Phi_\tau(u) := \Phi(u)\mathrm{e}^{\tau u^2}$ and $\Xi_\tau(z) := 8\int_0^\infty \Phi(u)\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u$ ($\partial_\tau \Xi_\tau \equiv -\frac{1}{4}\partial_z^2\Xi_\tau$, $\tau = 4t_{\mathrm{DBN}}$) to the transverse Chladni, Wigner, and dipole-cascade geometry of Riemann's Ξ-function. First, at $\tau = 0$ every simple zero $\gamma_k(\tau)$ moves with initial velocity $\frac{\mathrm{d}\gamma_k}{\mathrm{d}\tau}\Big|_{\tau=0} = \frac{\Xi''(\gamma_k)}{4\,\Xi'(\gamma_k)} \equiv \frac{1}{4}\kappa(\gamma_k) = \frac{1}{2}c_{\mathrm{par}}(\gamma_k)$ and every critical-line extremum $\tau_k(\tau)$ moves with velocity $\frac{\mathrm{d}\tau_k}{\mathrm{d}\tau}\Big|_{\tau=0} \equiv \frac{3}{4}\kappa_\Gamma(\tau_k)$, evaluating the principal-value Calogero–Moser sum in closed form as $\mathrm{P.V.}\sum_{\gamma_j\neq\gamma_k}(\gamma_k-\gamma_j)^{-1} \equiv \frac{1}{2}\kappa(\gamma_k)$. Second, in mirror coordinates $x = u+v, y = u-v$, the identity $u^2+v^2 \equiv \frac{1}{2}(x^2+y^2)$ yields the cross-term-free Gaussian Wigner factorization $G_{y,\tau}(x) \equiv \mathrm{e}^{\frac{\tau}{2}y^2}\mathrm{e}^{\frac{\tau}{2}x^2}G_y(x)$, preserving $|\mathrm{Im}(x)| < \frac{\pi}{4}$ and shifting the pitchfork coalescence threshold with velocity $\frac{\partial y_c}{\partial\tau}\Big|_{\tau=0} = \frac{\eta_c}{\partial_y L'_{y_c}(\eta_c)} > 0$. Third, every transverse dipole invariant of $\widetilde{W}_\tau(d,t) = \sum_{m=1}^\infty c_{2m-1}(t;\tau)d^{2m-2}$ obeys the universal PDE hierarchy $\partial_\tau c_{2m-1} + \frac{1}{8}\partial_t^2 c_{2m-1} \equiv \frac{m(2m+1)}{4}c_{2m+1}$, governing Lehmer-pair backward coalescence via $\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(t_{\mathrm{mid}}) = -2\frac{c_1(t_{\mathrm{mid}};0)}{c_3(t_{\mathrm{mid}};0)}(1+\mathcal{O}(\delta^2)) < 0$.

1. Introduction and Main Results

In the classical theory of the De Bruijn–Newman constant Λ initiated by Pólya [23], De Bruijn [6], and Newman [22], and developed quantitatively by Csordas, Norfolk, Smith, and Varga [2, 3, 4], Rodgers and Tao [26], and the Polymath 15 project [24], Riemann's completed zeta function $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u$ [25, 28] is embedded into a one-parameter family of entire functions of order one by multiplying the Jacobi–Riemann theta kernel $\Phi(u)$ by the Gaussian heat weight $\mathrm{e}^{\tau u^2}$:

$$\Phi_\tau(u) := \Phi(u)\,\mathrm{e}^{\tau u^2}, \qquad \Xi_\tau(z) := 8\int_0^\infty \Phi(u)\,\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u \qquad (z = t - \mathrm{i}d \in \mathbb{C},\ \tau \in \mathbb{R}). \tag{1.1}$$

In our canonical logarithmic coordinate u (related to the classical integration variable $w = 2u$ of [6, 22, 4, 26] by $\mathrm{e}^{t_{\mathrm{DBN}}w^2} = \mathrm{e}^{4t_{\mathrm{DBN}}u^2}$, so that $\tau = 4t_{\mathrm{DBN}}$ and the De Bruijn–Newman threshold is $\tau_{\mathrm{DBN}} = 4\Lambda$), the deformed function $\Xi_\tau(z)$ satisfies the backwards heat-flow partial differential equation $\partial_\tau\Xi_\tau(z) = -\frac{1}{4}\partial_z^2\Xi_\tau(z)$. While the longitudinal dynamics of the zeros $\gamma_k(\tau)$ along the critical line $d = 0$ have been extensively studied via the infinite Calogero–Moser / Dyson particle system [4, 26, 24] and Lehmer-pair lower bounds [11, 3], three structural questions connecting the

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longitudinal heat flow τ to the two-dimensional transverse geometry of the critical strip $d = \sigma - \frac{1}{2} > 0$ [27, 7, 10] have remained open:

(1) *Geometric duality between longitudinal heat-flow velocity and transverse Chladni nodal curvature.* When the real and imaginary nodal curves $R(d,t) = 0$ and $I(d,t) = 0$ of $\Xi(t - \mathrm{i}d) = R(d,t) + \mathrm{i}\, I(d,t)$ depart orthogonally from the critical line $d = 0$ into the strip $d > 0$ (Paper III [14]), how are their static parabolic curvatures $\kappa(\gamma_k)$ and $\kappa_\Gamma(\tau_k)$ related to the dynamical De Bruijn–Newman velocities $\gamma_k'(0)$ and $\tau_k'(0)$, and does this relation evaluate the conditionally convergent infinite Calogero–Moser sum in closed form?

(2) *Exact Wigner phase-space factorization under Gaussian deformation.* In the Wigner representation $W_\tau(d,t) = 8\int_0^\infty y \sinh(2dy)\, I_\tau(y,2t)\,\mathrm{d}y$ [29, 8] (Paper I [12]), does the Gaussian weight $\mathrm{e}^{\tau(u^2+v^2)}$ introduce a mixed xy cross-term in mirror coordinates $(x = u+v, y = u-v)$, or does it factor cleanly into an outer hyperbolic shift weight and an inner holomorphic contour kernel preserving the pitchfork saddle-point coalescence in $|\mathrm{Im}(x)| < \frac{\pi}{4}$?

(3) *Exact evolution hierarchy for the transverse dipole invariants $c_{2m-1}(t;\tau)$.* In Papers II and IV [13, 15], the parity-reduced Wronskian $\widetilde{W}_\tau(d,t) := W_\tau(d,t)/d = \sum_{m=1}^\infty c_{2m-1}(t;\tau) d^{2m-2}$ was shown to obey a universal square-cascade at multiple zeros and Lehmer pairs. What is the exact partial differential equation governing the evolution of every dipole invariant $c_{2m-1}(t;\tau)$ under τ?

Building upon the frameworks established in Papers I–X of this series [12, 13, 14, 15, 16, 17, 18, 19, 20, 21], the present work (Paper XI) resolves these three questions through three exact analytical theorems.

Theorem 1.1 (De Bruijn–Newman Chladni Velocity Duality, Cross-Term-Free Gaussian Wigner Factorization, and Universal Dipole PDE Hierarchy)**.** *Let $z = t - \mathrm{i}d \in \mathbb{C}$ and $\tau \in \mathbb{R}$, let $\Phi_\tau(u) := \Phi(u)\mathrm{e}^{\tau u^2}$, and let $\Xi_\tau(z) := 8\int_0^\infty \Phi(u)\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u = R_\tau(d,t) + \mathrm{i}\, I_\tau(d,t)$ with parity-reduced transverse Wronskian $\widetilde{W}_\tau(d,t) := \frac{1}{2d}\partial_d|\Xi_\tau(t-\mathrm{i}d)|^2 = \sum_{m=1}^\infty c_{2m-1}(t;\tau)d^{2m-2}$. Then the following three structural laws hold:*

(i) ***Backwards Heat-Flow PDE, Parabolic Chladni Curvature Duality, and Closed-Form Dyson Sum.*** *The function $\Xi_\tau(z)$ satisfies $\partial_\tau \Xi_\tau(z) \equiv -\frac{1}{4}\partial_z^2 \Xi_\tau(z)$ on $\mathbb{C}\times\mathbb{R}$. At $\tau = 0$, every simple real zero $\gamma_k(\tau)$ ($\Xi(\gamma_k) = 0, \Xi'(\gamma_k) \neq 0$) and every non-degenerate critical-line extremum $\tau_k(\tau)$ ($\Xi'(\tau_k) = 0, \Xi''(\tau_k) \neq 0$) move with exact initial velocities*

$$\left.\frac{\mathrm{d}\gamma_k}{\mathrm{d}\tau}\right|_{\tau=0} = \frac{\Xi''(\gamma_k)}{4\,\Xi'(\gamma_k)} \equiv \frac{1}{4}\kappa(\gamma_k) = \frac{1}{2}c_{\mathrm{par}}(\gamma_k), \qquad \left.\frac{\mathrm{d}\tau_k}{\mathrm{d}\tau}\right|_{\tau=0} = \frac{\Xi'''(\tau_k)}{4\,\Xi''(\tau_k)} \equiv \frac{3}{4}\kappa_\Gamma(\tau_k), \tag{1.2}$$

where $\kappa(\gamma_k) := t''_{\mathcal{R}_k}(0)$ and $\kappa_\Gamma(\tau_k) := \tau''_{\Gamma_k}(0)$ are the parabolic Chladni curvatures of $R(d,t) = 0$ and $I(d,t) = 0$ at $d = 0$, and the principal-value Calogero–Moser / Dyson interaction sum is evaluated in closed form by

$$\begin{aligned}\mathrm{P.V.}\sum_{\gamma_j\neq\gamma_k}\frac{1}{\gamma_k-\gamma_j} &:= \frac{1}{2\gamma_k} + \sum_{\substack{m=1\\ m\neq k}}^{\infty}\frac{2\gamma_k}{\gamma_k^2-\gamma_m^2} \\ &\equiv \frac{\Xi''(\gamma_k)}{2\,\Xi'(\gamma_k)} = \frac{1}{2}\kappa(\gamma_k) = -\frac{1}{2}\mathrm{Im}\left(2g_1\left(\tfrac{1}{2}+\mathrm{i}\gamma_k\right) + \frac{\zeta''(\frac{1}{2}+\mathrm{i}\gamma_k)}{\zeta'(\frac{1}{2}+\mathrm{i}\gamma_k)}\right).\end{aligned} \tag{1.3}$$

(ii) ***Cross-Term-Free Gaussian Wigner Factorization and Pitchfork Coalescence Shift.*** *In mirror coordinates $x = u+v$ and $y = u-v$, the identity $u^2+v^2 \equiv \frac{1}{2}(x^2+y^2)$ yields the cross-term-free factorization $G_{y,\tau}(x) := \Phi_\tau(\frac{x+y}{2})\Phi_\tau(\frac{x-y}{2}) \equiv \mathrm{e}^{\frac{\tau}{2}y^2}\mathrm{e}^{\frac{\tau}{2}x^2}G_y(x)$ and Wigner correlation PDE $\partial_\tau I_\tau(y,\omega) \equiv \frac{1}{2}(y^2 - \partial_\omega^2)I_\tau(y,\omega) \equiv \frac{1}{8}((2y)^2 - \partial_t^2)I_\tau(y,2t)$ ($\omega = 2t$), preserving the holomorphic strip $|\mathrm{Im}(x)| < \frac{\pi}{4}$ for all $\tau \in \mathbb{R}$ and shifting the pitchfork coalescence threshold*

$y_c(\omega;\tau)$ with strictly positive initial velocity

$$\left.\frac{\partial y_c(\omega;\tau)}{\partial\tau}\right|_{\tau=0} = \frac{\eta_c(\omega;0)}{\partial_y L'_{y_c}(\eta_c(\omega;0))} > 0. \tag{1.4}$$

(iii) ***Universal Transverse Dipole Evolution PDE Hierarchy and Lehmer Coalescence Law.*** *The parity-reduced Wronskian $\widetilde{W}_\tau(d,t)$ satisfies the fourth-order radial wave-heat PDE $\partial_\tau\widetilde{W}_\tau + \frac{1}{8}(\partial_t^2 - \partial_d^2 - \frac{2}{d}\partial_d)\widetilde{W}_\tau \equiv 0$ together with $\partial_\tau\widetilde{W}_\tau + \frac{1}{4}\partial_t^2\widetilde{W}_\tau \equiv \frac{1}{8}(\partial_t^2 + \partial_d^2 + \frac{2}{d}\partial_d)\widetilde{W}_\tau \equiv \frac{1}{2}\widetilde{W}_{1,2;\tau}(d,t)$ ($\widetilde{W}_{1,2;\tau} := \frac{1}{2d}\partial_d|\Xi'_\tau(t-\mathrm{id})|^2$), and every transverse dipole invariant $c_{2m-1}(t;\tau)$ obeys the universal PDE hierarchy*

$$\frac{\partial c_{2m-1}}{\partial\tau}(t;\tau) + \frac{1}{8}\frac{\partial^2 c_{2m-1}}{\partial t^2}(t;\tau) \equiv \frac{m(2m+1)}{4}\,c_{2m+1}(t;\tau) \qquad (\forall\, m\geq 1). \tag{1.5}$$

At any m-fold coalescence point t_0 ($m\geq 2$), $\partial_\tau c_{2m-3}(t_0;\tau_{\mathrm{coal}}) \equiv \frac{m}{4}c_{2m-1}(t_0;\tau_{\mathrm{coal}}) = \frac{m^2}{4(m!)^2}(\partial_t^m\Xi_{\tau_{\mathrm{coal}}}(t_0))^2 > 0$, and across any closely spaced Lehmer pair $t_\pm = t_{\mathrm{mid}} \pm \delta$ ($\delta>0$), $\left.\frac{\partial c_1}{\partial\tau}\right|_{(t_{\mathrm{mid}},0)} \equiv \frac{1}{4}(\Xi''(t_{\mathrm{mid}}))^2 + \frac{1}{4}\Xi(t_{\mathrm{mid}})\Xi^{(4)}(t_{\mathrm{mid}}) > 0$, recovering the backward coalescence threshold $\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(t_{\mathrm{mid}}) = -2\frac{c_1(t_{\mathrm{mid}};0)}{c_3(t_{\mathrm{mid}};0)}(1+\mathcal{O}(\delta^2)) = -2\delta^2(1+\mathcal{O}(\delta^2)) < 0$.

The canonical quantitative constants governing Theorem 1.1 are given explicitly by

$$\begin{aligned}
&\kappa(\gamma_1) = -1.1592611550\ldots, && \left.\frac{\mathrm{d}\gamma_1}{\mathrm{d}\tau}\right|_{\tau=0} = \frac{1}{4}\kappa(\gamma_1) = -0.2898152887\ldots,\\
&\left.\frac{\mathrm{d}(\gamma_{10}-\gamma_9)}{\mathrm{d}\tau}\right|_{\tau=0} = +0.3866151652\cdots > 0, && \left.\frac{\partial y_c(2\gamma_1;\tau)}{\partial\tau}\right|_{\tau=0} = +0.0125362327\cdots > 0,\\
&\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(\gamma_9,\gamma_{10}) = -1.34840873\ldots, && \tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(\gamma_{6709},\gamma_{6710}) = -7.096664\times 10^{-4}.
\end{aligned} \tag{1.6}$$

1.1. **Proof overview.** The proof of Theorem 1.1 combines three quantitative ingredients developed in Sections 2–4:

(1) *Harmonic Cauchy–Riemann curvature versus backwards heat-flow differentiation (Section 2).* Because $\Phi(u) \ll \mathrm{e}^{9|u|}\exp(-\pi\mathrm{e}^{4|u|})$ decays double-exponentially, differentiating $\Xi_\tau(z)$ under the integral sign yields $\partial_\tau\Xi_\tau = -\frac{1}{4}\partial_z^2\Xi_\tau$. Since the two-dimensional Laplace equation $\partial_d^2 R = -\partial_t^2 R$ on the critical line $d=0$ and the one-dimensional backwards heat equation $\partial_\tau\Xi_\tau = -\frac{1}{4}\partial_t^2\Xi_\tau$ at $\tau=0$ are governed by the identical operator $-\partial_t^2$, implicit differentiation of $\Xi_\tau(\gamma_k(\tau)) = 0$ and $\partial_t\Xi_\tau(\tau_k(\tau)) = 0$ directly identifies $\gamma_k'(0)$ and $\tau_k'(0)$ with the Chladni curvatures $\frac{1}{4}\kappa(\gamma_k)$ and $\frac{3}{4}\kappa_\Gamma(\tau_k)$, while matching the regular part of $\Xi'(z)/\Xi(z)$ at $z=\gamma_k$ evaluates the Calogero–Moser sum (1.3).

(2) *Parallelogram diagonalization in mirror coordinates and implicit pitchfork perturbation (Section 3).* Under the orthogonal rotation $u = \frac{x+y}{2}, v = \frac{x-y}{2}$, the quadratic form $u^2+v^2 = \frac{1}{2}(x^2+y^2)$ is diagonal. Consequently, $G_{y,\tau}(x) = \mathrm{e}^{\frac{\tau}{2}y^2}\mathrm{e}^{\frac{\tau}{2}x^2}G_y(x)$ factors without an xy cross-term, preserving the double-exponential closure of the Cauchy contour box in $|\mathrm{Im}(x)| < \frac{\pi}{4}$. Differentiating the simultaneous pitchfork bifurcation system $L'_{y_c}(\eta_c) - \tau\eta_c = \omega$ and $L''_{y_c}(\eta_c) = \tau$ at $\tau=0$, the term $L''_{y_c}(\eta_c)\,\partial_\tau\eta_c$ vanishes identically by the coalescence condition $L''_{y_c}(\eta_c) = 0$, isolating (1.4).

(3) *Radial wave-heat operator cancellation and Taylor moment hierarchy (Section 4).* For $M_\tau(d,t) := |\Xi_\tau(t-\mathrm{id})|^2$, the gradient squares $+2|\Xi'_\tau|^2$ in $\partial_t^2 M_\tau$ and $\partial_d^2 M_\tau$ cancel identically in $(\partial_t^2 - \partial_d^2)M_\tau = -8\,\partial_\tau M_\tau$ and add in $(\partial_t^2+\partial_d^2)M_\tau = 4|\Xi'_\tau|^2$. Applying $\frac{1}{2d}\partial_d$ and expanding $\widetilde{W}_\tau(d,t) = \sum_{m=1}^\infty c_{2m-1}(t;\tau)d^{2m-2}$ yields the coupled hierarchy (1.5), which we combine with the Weierstrass normal form at Lehmer pairs to establish the backward coalescence law.

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ with $d = \sigma - \frac{1}{2} \in \mathbb{R}$ and $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$. Riemann's completed zeta function is $\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s) = G(s)\zeta(s)$, where $G(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})$ has logarithmic derivative $g_1(s) := \frac{G'(s)}{G(s)} = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\ln\pi + \frac{1}{2}\psi(\frac{s}{2})$. For $\tau \in \mathbb{R}$, the De Bruijn–Newman deformed function is $\Xi_\tau(z) := 8\int_0^\infty \Phi(u)\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u = R_\tau(d,t) + \mathrm{i}\,I_\tau(d,t)$, where $\Phi(u) := \sum_{m=1}^\infty (2\pi^2 m^4 \mathrm{e}^{9u} - 3\pi m^2 \mathrm{e}^{5u})\exp(-\pi m^2 \mathrm{e}^{4u})$ satisfies $\Phi(-u) = \Phi(u) > 0$. The transverse Wronskian is $W_\tau(d,t) := \frac{1}{2}\partial_d|\Xi_\tau(t-\mathrm{i}d)|^2 = R_\tau\,\partial_d R_\tau + I_\tau\,\partial_d I_\tau = \mathrm{Im}(\overline{\Xi_\tau(t-\mathrm{i}d)}\,\partial_z\Xi_\tau(t-\mathrm{i}d))$, and the parity-reduced Wronskian is $\widetilde{W}_\tau(d,t) := W_\tau(d,t)/d$. All numerical values are computed at 60-decimal working precision (`mpmath.mp.dps = 60`) and certified via 35-decimal directed-rounding interval arithmetic (`mpmath.iv.dps = 35`) in `scripts/verify_paper11_debruijn_newman_chladni.py`.

2. Backwards Heat-Flow PDE, Parabolic Chladni Zero-Curve Curvature $\kappa(\gamma_k)$, and Closed-Form Calogero–Moser Sum

Before differentiating $\Xi_\tau(z)$ with respect to τ and z, we establish the absolute and locally uniform convergence of all integral derivatives across $\mathbb{C}\times\mathbb{R}$. For real $u \geq 0$, the Jacobi–Riemann kernel satisfies $0 < \Phi(u) \leq C_0\,\mathrm{e}^{9u}\exp(-\pi\mathrm{e}^{4u})$ with $C_0 = 2\pi^2\sum_{m=1}^\infty m^4\mathrm{e}^{-\pi(m^2-1)} < 20$. For any bounded subset $K \subset \mathbb{C}\times\mathbb{R}$ with $|z| \leq R$ and $|\tau| \leq T$, and for any integers $j,k \geq 0$, the differentiated integrand satisfies

$$\left|\partial_\tau^j\partial_z^k\left[\Phi(u)\,\mathrm{e}^{\tau u^2}\cos(2zu)\right]\right| \leq C_0\,(2u)^{2j+k}\exp\!\big(9u + Tu^2 + 2Ru - \pi\mathrm{e}^{4u}\big) \qquad (u \geq 0). \tag{2.1}$$

Because $\pi\mathrm{e}^{4u}$ grows exponentially while $Tu^2 + (2R+9)u$ grows only quadratically, the right-hand side of (2.1) is integrable on $[0,\infty)$ uniformly on K. By Lebesgue's dominated convergence theorem and Morera's theorem, $\Xi_\tau(z)$ is an even entire function of z for every $\tau \in \mathbb{R}$ and is real-analytic in $(z,\tau) \in \mathbb{C}\times\mathbb{R}$, permitting termwise differentiation under the integral sign to all orders.

Theorem 2.1 (Backwards Heat-Flow PDE and Parabolic Chladni Curvature Duality)**.** *Let $\Phi_\tau(u) := \Phi(u)\mathrm{e}^{\tau u^2}$ and $\Xi_\tau(z) := 8\int_0^\infty \Phi(u)\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u$. Then:*

(i) ***Exact Backwards Heat-Flow PDE.*** *For all $(z,\tau) \in \mathbb{C}\times\mathbb{R}$, $\Xi_\tau(z)$ satisfies the backwards heat equation*

$$\frac{\partial\Xi_\tau}{\partial\tau}(z) \equiv -\frac{1}{4}\frac{\partial^2\Xi_\tau}{\partial z^2}(z) \qquad \left(\textit{equivalently } \frac{\partial\Xi_\tau}{\partial t_{\mathrm{DBN}}}(z) \equiv -\frac{\partial^2\Xi_\tau}{\partial z^2}(z) \textit{ for } t_{\mathrm{DBN}} = \frac{\tau}{4}\right). \tag{2.2}$$

(ii) ***Initial Zero Velocity and Parabolic Chladni Zero-Curve Curvature $\kappa(\gamma_k)$.*** *Let $\gamma_k(\tau)$ be a simple zero of Ξ_τ continuing the k-th simple critical-line zero $\gamma_k = \gamma_k(0)$ of $\Xi(t)$, and let $\mathcal{R}_k : R(d, t_{\mathcal{R}_k}(d)) \equiv 0$ be the real Chladni nodal curve departing orthogonally into the critical strip $d > 0$ with Taylor expansion $t_{\mathcal{R}_k}(d) = \gamma_k + c_{\mathrm{par}}(\gamma_k)d^2 + c_{\mathrm{quart}}(\gamma_k)d^4 + \mathcal{O}(d^6)$ and parabolic curvature $\kappa(\gamma_k) := t''_{\mathcal{R}_k}(0) = 2c_{\mathrm{par}}(\gamma_k) = \frac{\Xi''(\gamma_k)}{\Xi'(\gamma_k)}$. Then for every τ where $\partial_z\Xi_\tau(\gamma_k(\tau)) \neq 0$, $\gamma_k'(\tau) = \frac{\partial_z^2\Xi_\tau(\gamma_k(\tau))}{4\,\partial_z\Xi_\tau(\gamma_k(\tau))}$, and at $\tau = 0$:*

$$\left.\frac{\mathrm{d}\gamma_k}{\mathrm{d}\tau}\right|_{\tau=0} = \frac{\Xi''(\gamma_k)}{4\,\Xi'(\gamma_k)} \equiv \frac{1}{4}\kappa(\gamma_k) \equiv \frac{1}{2}c_{\mathrm{par}}(\gamma_k) \equiv -\frac{1}{4}\mathrm{Im}\left(2g_1\left(\tfrac{1}{2}+\mathrm{i}\gamma_k\right) + \frac{\zeta''(\frac{1}{2}+\mathrm{i}\gamma_k)}{\zeta'(\frac{1}{2}+\mathrm{i}\gamma_k)}\right), \tag{2.3}$$

while the second-order heat-flow acceleration at $\tau = 0$ is connected to the quartic Chladni coefficient $c_{\mathrm{quart}}(\gamma_k) = \frac{\Xi'''(\gamma_k)\kappa(\gamma_k)}{4\,\Xi'(\gamma_k)} - \frac{\Xi^{(4)}(\gamma_k)}{24\,\Xi'(\gamma_k)} - \frac{1}{8}\kappa(\gamma_k)^3$ by

$$\left.\frac{\mathrm{d}^2\gamma_k}{\mathrm{d}\tau^2}\right|_{\tau=0} = \frac{\Xi'''(\gamma_k)\,\kappa(\gamma_k)}{8\,\Xi'(\gamma_k)} - \frac{\Xi^{(4)}(\gamma_k)}{16\,\Xi'(\gamma_k)} - \frac{1}{16}\kappa(\gamma_k)^3 \equiv \frac{3}{2}c_{\mathrm{quart}}(\gamma_k) - \frac{\Xi'''(\gamma_k)\,\kappa(\gamma_k)}{4\,\Xi'(\gamma_k)} + \frac{1}{8}\kappa(\gamma_k)^3. \tag{2.4}$$

(iii) ***Initial Extremum Velocity and Imaginary Chladni Wall Curvature*** $\kappa_\Gamma(\tau_k)$. *Let* $\tau_k(\tau) \in (\gamma_k(\tau), \gamma_{k+1}(\tau))$ *be a non-degenerate critical-line extremum satisfying* $\partial_t \Xi_\tau(\tau_k(\tau)) = 0$ *and* $\partial_t^2 \Xi_\tau(\tau_k(\tau)) \neq 0$, *and let* $\Gamma_k : I(d, \tau_{\Gamma_k}(d)) \equiv 0$ *be the imaginary Chladni chamber wall at* $\tau = 0$ *with parabolic wall curvature* $\kappa_\Gamma(\tau_k) := \tau''_{\Gamma_k}(0) = \frac{\Xi'''(\tau_k)}{3\,\Xi''(\tau_k)}$. *Then at* $\tau = 0$*:*

$$\left.\frac{\mathrm{d}\tau_k}{\mathrm{d}\tau}\right|_{\tau=0} = \frac{\Xi'''(\tau_k)}{4\,\Xi''(\tau_k)} \equiv \frac{3}{4}\kappa_\Gamma(\tau_k). \tag{2.5}$$

Proof. (i) Differentiating (1.1) under the integral sign via (2.1) gives $\partial_\tau \Xi_\tau(z) = 8\int_0^\infty u^2\Phi(u)\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u$ and $\partial_z^2\Xi_\tau(z) = 8\int_0^\infty(-4u^2)\Phi(u)\mathrm{e}^{\tau u^2}\cos(2zu)\,\mathrm{d}u = -4\,\partial_\tau\Xi_\tau(z)$, proving (2.2).

(ii) Since $\Xi_\tau(\gamma_k(\tau)) \equiv 0$ and $\partial_z\Xi_\tau(\gamma_k(\tau)) \neq 0$, the implicit function theorem guarantees that $\tau \mapsto \gamma_k(\tau)$ is real-analytic near $\tau = 0$, and differentiating $\Xi_\tau(\gamma_k(\tau)) = 0$ with respect to τ yields $\partial_\tau\Xi_\tau(\gamma_k(\tau)) + \partial_z\Xi_\tau(\gamma_k(\tau))\,\gamma_k'(\tau) = 0$. Substituting $\partial_\tau\Xi_\tau = -\frac{1}{4}\partial_z^2\Xi_\tau$ gives $\gamma_k'(\tau) = \frac{\partial_z^2\Xi_\tau(\gamma_k(\tau))}{4\,\partial_z\Xi_\tau(\gamma_k(\tau))}$. Meanwhile, on the critical strip $z = t - \mathrm{i}d$ at $\tau = 0$, expanding $R(d,t) = \mathrm{Re}\,\Xi(t - \mathrm{i}d) = \Xi(t) - \frac{d^2}{2}\Xi''(t) + \frac{d^4}{24}\Xi^{(4)}(t) + \mathcal{O}(d^6)$ along the real nodal curve $t_{\mathcal{R}_k}(d) = \gamma_k + c_{\mathrm{par}}d^2 + c_{\mathrm{quart}}d^4 + \mathcal{O}(d^6)$ where $\Xi(\gamma_k) = 0$ gives, at order d^2, $\Xi'(\gamma_k)c_{\mathrm{par}} - \frac{1}{2}\Xi''(\gamma_k) = 0$, so $\kappa(\gamma_k) = t''_{\mathcal{R}_k}(0) = 2c_{\mathrm{par}}(\gamma_k) = \frac{\Xi''(\gamma_k)}{\Xi'(\gamma_k)}$, establishing the first three equalities in (2.3). Next, writing $\Xi(z) = G(s)\zeta(s)$ with $s = \frac{1}{2} + \mathrm{i}z$ ($\frac{\mathrm{d}s}{\mathrm{d}z} = \mathrm{i}$) and $G'(s) = g_1(s)G(s)$, the chain rule gives $\Xi'(z) = \mathrm{i}(G'(s)\zeta(s) + G(s)\zeta'(s))$ and $\Xi''(z) = -(G''(s)\zeta(s) + 2G'(s)\zeta'(s) + G(s)\zeta''(s))$. Evaluating at $z = \gamma_k$ where $\zeta(s_k) = 0$ ($s_k = \frac{1}{2} + \mathrm{i}\gamma_k$) yields

$$\frac{\Xi''(\gamma_k)}{\Xi'(\gamma_k)} = -\frac{2G'(s_k)\zeta'(s_k) + G(s_k)\zeta''(s_k)}{\mathrm{i}\,G(s_k)\zeta'(s_k)} = \mathrm{i}\left(2g_1(s_k) + \frac{\zeta''(s_k)}{\zeta'(s_k)}\right) = -\mathrm{Im}\left(2g_1(s_k) + \frac{\zeta''(s_k)}{\zeta'(s_k)}\right),$$

where the last step uses that $\Xi''(\gamma_k)/\Xi'(\gamma_k)$ is real. Dividing by 4 gives the final equality in (2.3). For (2.4), expanding $\Xi_\tau(\gamma_k + v_1\tau + \frac{1}{2}v_2\tau^2) = 0$ ($v_1 = \gamma_k'(0), v_2 = \gamma_k''(0)$) using $\Xi_\tau(z) = \Xi(z) - \frac{\tau}{4}\Xi''(z) + \frac{\tau^2}{32}\Xi^{(4)}(z) + \mathcal{O}(\tau^3)$ gives, at order τ^2, $\frac{1}{2}\Xi' v_2 + \frac{1}{2}\Xi'' v_1^2 - \frac{1}{4}\Xi''' v_1 + \frac{1}{32}\Xi^{(4)} = 0$, which yields (2.4) upon comparing with the order-d^4 equation $\Xi' c_{\mathrm{quart}} + \frac{1}{2}\Xi'' c_{\mathrm{par}}^2 - \frac{1}{2}\Xi''' c_{\mathrm{par}} + \frac{1}{24}\Xi^{(4)} = 0$.

(iii) Differentiating the extremum condition $\partial_t\Xi_\tau(\tau_k(\tau)) \equiv 0$ with respect to τ and substituting $\partial_\tau\partial_t\Xi_\tau = -\frac{1}{4}\partial_t^3\Xi_\tau$ yields $\tau_k'(\tau) = \frac{\partial_t^3\Xi_\tau(\tau_k(\tau))}{4\,\partial_t^2\Xi_\tau(\tau_k(\tau))}$. At $\tau = 0$, the imaginary nodal curve $-I(d,t)/d = \Xi'(t) - \frac{d^2}{6}\Xi'''(t) + \mathcal{O}(d^4) = 0$ has $\kappa_\Gamma(\tau_k) = \tau''_{\Gamma_k}(0) = \frac{\Xi'''(\tau_k)}{3\,\Xi''(\tau_k)}$ (Paper III [14], Proposition 2.1(b); Paper X [21], Theorem 4.1(i)), which proves (2.5). □

Corollary 2.2 (Closed-Form Evaluation of the Principal-Value Calogero–Moser / Dyson Sum)**.** *Let* $\{\pm\gamma_m(\tau)\}_{m=1}^\infty$ *be the simple zeros of the even entire function* $\Xi_\tau(z)$ *of order* 1. *Then for every* $k \geq 1$, *the symmetric principal-value Calogero–Moser / Dyson interaction sum over all roots* $\gamma_j(\tau) \in \{\pm\gamma_m(\tau)\}_{m=1}^\infty \setminus \{+\gamma_k(\tau)\}$ *converges absolutely when paired by* $\pm\gamma_m(\tau)$ *and satisfies*

$$\mathrm{P.V.}\sum_{\gamma_j\neq\gamma_k}\frac{1}{\gamma_k(\tau) - \gamma_j(\tau)} := \frac{1}{2\gamma_k(\tau)} + \sum_{\substack{m=1\\ m\neq k}}^\infty \frac{2\gamma_k(\tau)}{\gamma_k(\tau)^2 - \gamma_m(\tau)^2} \equiv 2\,\frac{\mathrm{d}\gamma_k}{\mathrm{d}\tau}(\tau) = \frac{\partial_z^2\Xi_\tau(\gamma_k(\tau))}{2\,\partial_z\Xi_\tau(\gamma_k(\tau))}. \tag{2.6}$$

In particular, at $\tau = 0$, $\mathrm{P.V.}\sum_{\gamma_j\neq\gamma_k}\frac{1}{\gamma_k - \gamma_j} \equiv \frac{\Xi''(\gamma_k)}{2\,\Xi'(\gamma_k)} \equiv \frac{1}{2}\kappa(\gamma_k) \equiv c_{\mathrm{par}}(\gamma_k)$.

Proof. Since $\Xi_\tau(z)$ is an even entire function of order 1, Hadamard's factorization theorem gives $\Xi_\tau(z) = \Xi_\tau(0)\prod_{m=1}^\infty(1 - \frac{z^2}{\gamma_m(\tau)^2})$. Taking the logarithmic derivative and separating the $m = k$ term yields

$$\frac{\partial_z\Xi_\tau(z)}{\Xi_\tau(z)} - \frac{1}{z - \gamma_k(\tau)} = \frac{1}{z + \gamma_k(\tau)} + \sum_{\substack{m=1\\ m\neq k}}^\infty \frac{2z}{z^2 - \gamma_m(\tau)^2},$$

TABLE 1. Part A: 60-decimal verification of the De Bruijn–Newman velocity duality (2.3), the closed-form Calogero–Moser / Dyson sum (2.6), and the Archimedean–Zeta identity across nontrivial zeros γ_k. Part B: Verification of the dual extremum wall law (2.5) across the first six critical-line extrema $\tau_k(0) \in (\gamma_k, \gamma_{k+1})$.

Part A: Zero Velocity $\gamma_k'(0) \equiv \frac{1}{4}\kappa(\gamma_k)$ and Closed-Form Dyson Sum						Part B: Extremum Wall $\tau_k'(0) \equiv \frac{3}{4}\kappa_\Gamma(\tau_k)$			
k	γ_k	$\kappa(\gamma_k)$	P.V. $\sum \frac{1}{\gamma_k-\gamma_j}$	$\gamma_k'(0) = \frac{1}{4}\kappa$	Arch–Zeta	k	$\tau_k(0)$	$\kappa_\Gamma(\tau_k)$	$\tau_k'(0) = \frac{3}{4}\kappa_\Gamma$
1	14.13472514	−1.15926116	−0.57963058	−0.28981529	3.89×10^{-62}	1	15.58570859	−0.37992155	−0.28494116
2	21.02203964	−1.50999100	−0.75499550	−0.37749775	2.72×10^{-61}	2	22.09797728	−0.47940252	−0.35955189
3	25.01085758	−1.17166632	−0.58583316	−0.29291658	3.89×10^{-62}	3	26.27224736	−0.40605036	−0.30453777
4	30.42487613	−1.91084251	−0.95542126	−0.47771063	1.56×10^{-61}	4	31.23179587	−0.56716042	−0.42537032
5	32.93506159	−0.93089092	−0.46544546	−0.23272273	5.06×10^{-61}	5	34.19331027	−0.36989105	−0.27741829
9	48.00515088	−2.39696888	−1.19848444	−0.59924222	7.78×10^{-61}	6	38.49824076	−0.54150874	−0.40613155
10	49.77383248	−0.85050822	−0.42525411	−0.21262705	6.42×10^{-61}	Gap vel. $\Delta_9'(0) = +0.38661517 > 0$			
13	59.34704400	−2.25187946	−1.12593973	−0.56296987	5.45×10^{-61}	Gap vel. $\Delta_{13}'(0) = +0.49064704 > 0$			
14	60.83177852	−0.28929129	−0.14464565	−0.07232282	2.33×10^{-61}	Ratio err $\lvert\frac{4\tau_k'(0)}{3\kappa_\Gamma} - 1\rvert \le 10^{-59}$			

where the series on the right converges absolutely and uniformly on compact sets avoiding $\{\pm\gamma_m(\tau)\}_{m\neq k}$ because $\sum_{m=1}^{\infty} |\gamma_m(\tau)|^{-2} < \infty$. On the other hand, expanding $\Xi_\tau(z) = \Xi_\tau'(\gamma_k)(z-\gamma_k) + \frac{1}{2}\Xi_\tau''(\gamma_k)(z-\gamma_k)^2 + \mathcal{O}((z-\gamma_k)^3)$ as $z \to \gamma_k(\tau)$ gives $\lim_{z\to\gamma_k(\tau)}(\frac{\partial_z\Xi_\tau(z)}{\Xi_\tau(z)} - \frac{1}{z-\gamma_k(\tau)}) = \frac{\partial_z^2\Xi_\tau(\gamma_k(\tau))}{2\,\partial_z\Xi_\tau(\gamma_k(\tau))}$. Combining this limit with Theorem 2.1(ii) yields (2.6) (reproducing the Csordas–Smith–Varga / Rodgers–Tao / Polymath 15 ODE [4, 26, 24] in the scaling $\tau = 4t_{\mathrm{DBN}}$). □

3. Cross-Term-Free Gaussian Wigner Factorization and Saddle Pitchfork Shift

We next examine how the De Bruijn–Newman heat flow deforms the Wigner phase-space distribution [29, 8] and the pitchfork saddle-point coalescence established in Paper I [12]. Substituting $\Phi_\tau(u) = \Phi(u)e^{\tau u^2}$ into the double integral for $|\Xi_\tau(t - \mathrm{i}d)|^2$ and passing to mirror phase-space coordinates $x = u + v \in (-\infty, \infty)$ and $y = u - v \in [0, \infty)$ ($u = \frac{x+y}{2}, v = \frac{x-y}{2}$, Jacobian $\frac{1}{2}$) yields

$$\widetilde{W}_\tau(d,t) = 16\int_0^\infty y^2\,\mathrm{sinc}_{\mathrm{hyp}}(2dy)\,I_\tau(y,2t)\,\mathrm{d}y, \qquad I_\tau(y,\omega) := \int_{-\infty}^{\infty} G_{y,\tau}(x)\cos(\omega x)\,\mathrm{d}x, \tag{3.1}$$

where $\mathrm{sinc}_{\mathrm{hyp}}(w) := \frac{\sinh w}{w}$ and $G_{y,\tau}(x) := \Phi_\tau(\frac{x+y}{2})\Phi_\tau(\frac{x-y}{2})$.

Before shifting the x-contour of $I_\tau(y,\omega)$ into the complex strip $\mathcal{S}_{\pi/4} := \{x = \xi + \mathrm{i}\eta \in \mathbb{C} : |\eta| < \frac{\pi}{4}\}$, we verify that the holomorphic strip and vertical Cauchy wall decay of Paper I remain invariant for all $\tau \in \mathbb{R}$. Since $x \mapsto e^{\frac{\tau}{2}x^2}$ is an entire function of x, $G_{y,\tau}(x)$ is holomorphic on the same strip $\mathcal{S}_{\pi/4}$ as $G_y(x)$. On the vertical boundary segments $x = \pm L + \mathrm{i}\eta_0$ ($0 \le \eta_0 \le \eta < \frac{\pi}{4}$) as $L \to \infty$, we have $|e^{\frac{\tau}{2}(\pm L+\mathrm{i}\eta_0)^2}| = \exp(\frac{\tau}{2}(L^2 - \eta_0^2))$, while by Proposition 3.2 of Paper I [12], $|G_y(\pm L + \mathrm{i}\eta_0)| \le C_2\exp(9L - 2\pi\cosh(2y)e^{2L}\cos(2\eta_0))$. Because $\cos(2\eta_0) \ge \cos(2\eta) > 0$ for $|\eta| < \frac{\pi}{4}$, the double-exponential decay $-2\pi\cosh(2y)e^{2L}\cos(2\eta)$ dominates the quadratic exponent $+\frac{|\tau|}{2}L^2$ for every fixed $\tau \in \mathbb{R}$:

$$\left|G_{y,\tau}(\pm L + \mathrm{i}\eta_0)\,e^{\mathrm{i}\omega(\pm L+\mathrm{i}\eta_0)}\right| \le C_2\,e^{\frac{\tau}{2}(y^2-\eta_0^2)}\exp\!\left(9L + \frac{|\tau|}{2}L^2 - 2\pi\cosh(2y)\,e^{2L}\cos(2\eta)\right) \xrightarrow{L\to\infty} 0. \tag{3.2}$$

By Cauchy's theorem, the horizontal contour shift $x = \xi + \mathrm{i}\eta$ ($|\eta| < \frac{\pi}{4}$) is strictly justified for all $\tau \in \mathbb{R}$.

Theorem 3.1 (Cross-Term-Free Gaussian Wigner Factorization and Saddle Pitchfork Shift)**.** *Let $x = u + v \in \mathcal{S}_{\pi/4}$, $y = u - v \ge 0$, $\omega = 2t > 0$, and $\tau \in \mathbb{R}$. Then:*

(i) ***Cross-Term-Free Gaussian Factorization and Wigner Correlation PDE.*** *Because* $u^2+v^2=(\frac{x+y}{2})^2+(\frac{x-y}{2})^2\equiv\frac{1}{2}(x^2+y^2)$ *has zero* xy *cross-term, the deformed mirror kernel* $G_{y,\tau}(x)$ *and Wigner correlation kernel* $I_\tau(y,\omega)$ *satisfy the exact identities*

$$\boxed{\begin{aligned} G_{y,\tau}(x) &:= \Phi_\tau\Big(\frac{x+y}{2}\Big)\,\Phi_\tau\Big(\frac{x-y}{2}\Big) \equiv \mathrm{e}^{\frac{\tau}{2}y^2}\,\mathrm{e}^{\frac{\tau}{2}x^2}\,G_y(x),\\ \partial_\tau I_\tau(y,\omega) &\equiv \frac{1}{2}(y^2-\partial_\omega^2)I_\tau(y,\omega) \equiv \frac{1}{8}((2y)^2-\partial_t^2)I_\tau(y,2t), \end{aligned}} \tag{3.3}$$

so that $I_\tau(y,\omega)\equiv \mathrm{e}^{\frac{\tau}{2}y^2}\widetilde{I}_\tau(y,\omega)$ *with* $\widetilde{I}_\tau(y,\omega):=\int_{-\infty}^{\infty}\mathrm{e}^{\frac{\tau}{2}x^2}G_y(x)\cos(\omega x)\,\mathrm{d}x$.

(ii) ***Preservation of Hermitian Mirror Symmetry in*** $|\mathrm{Im}(x)|<\frac{\pi}{4}$**.** *For every* $\tau\in\mathbb{R}$ *and every* $x=\xi+\mathrm{i}\eta\in\mathcal{S}_{\pi/4}$, $G_{y,\tau}(-\xi+\mathrm{i}\eta)=\overline{G_{y,\tau}(\xi+\mathrm{i}\eta)}$, *and along the imaginary axis* $\xi=0$, $G_{y,\tau}(\mathrm{i}\eta)=\mathrm{e}^{\frac{\tau}{2}(y^2-\eta^2)}|\Phi(\frac{y+\mathrm{i}\eta}{2})|^2>0$ *is real and strictly positive.*

(iii) ***Exact Deformed Pitchfork Coalescence System and Positive Threshold Velocity.*** *Let* $L_y(\eta):=\ln G_y(\mathrm{i}\eta)=2\ln|\Phi(\frac{y+\mathrm{i}\eta}{2})|$ *for* $\eta\in(0,\frac{\pi}{4})$. *Under the heat deformation* τ, *the complex saddle points of* $\Psi_{y,\omega,\tau}(x):=\ln G_{y,\tau}(x)+\mathrm{i}\omega x$ *coalesce on the imaginary axis* $x=\mathrm{i}\eta_c(\omega;\tau)$ *at* $y=y_c(\omega;\tau)$ *determined by the simultaneous bifurcation system*

$$L'_{y_c}(\eta_c(\omega;\tau))-\tau\,\eta_c(\omega;\tau)=\omega,\qquad L''_{y_c}(\eta_c(\omega;\tau))=\tau. \tag{3.4}$$

At $\tau=0$, *where* $\partial_y L'_{y_c}(\eta_c(\omega;0))>0$ *and* $L'''_{y_c}(\eta_c(\omega;0))<0$, *the coalescence threshold* $y_c(\omega;\tau)$ *shifts with strictly positive initial velocity*

$$\left.\frac{\partial y_c(\omega;\tau)}{\partial\tau}\right|_{\tau=0}=\frac{\eta_c(\omega;0)}{\partial_y L'_{y_c}(\eta_c(\omega;0))}>0,\qquad \left.\frac{\partial\eta_c(\omega;\tau)}{\partial\tau}\right|_{\tau=0}=\frac{1-(\partial_y L''_{y_c}(\eta_c))\,\partial_\tau y_c(\omega;0)}{L'''_{y_c}(\eta_c(\omega;0))}<0. \tag{3.5}$$

Proof. (i) Expanding $(\frac{x+y}{2})^2+(\frac{x-y}{2})^2=\frac{1}{4}(x^2+2xy+y^2+x^2-2xy+y^2)=\frac{1}{2}(x^2+y^2)$ gives $\Phi_\tau(\frac{x+y}{2})\Phi_\tau(\frac{x-y}{2})=\mathrm{e}^{\tau(u^2+v^2)}\Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})=\mathrm{e}^{\frac{\tau}{2}y^2}\mathrm{e}^{\frac{\tau}{2}x^2}G_y(x)$. Differentiating $I_\tau(y,\omega)=\int_{-\infty}^{\infty}\mathrm{e}^{\frac{\tau}{2}(x^2+y^2)}G_y(x)\cos(\omega x)\,\mathrm{d}x$ under the integral sign with respect to τ brings down $\frac{1}{2}(y^2+x^2)$, while $\partial_\omega^2\cos(\omega x)=-x^2\cos(\omega x)$ and $\partial_t^2\cos(2tx)=-4x^2\cos(2tx)$ $(\omega=2t)$, which yields (3.3).

(ii) By Proposition 3.3 of Paper I [12], $G_y(-\xi+\mathrm{i}\eta)=\overline{G_y(\xi+\mathrm{i}\eta)}$ and $G_y(\mathrm{i}\eta)=|\Phi(\frac{y+\mathrm{i}\eta}{2})|^2>0$. Since $\mathrm{e}^{\frac{\tau}{2}(-\xi+\mathrm{i}\eta)^2}=\mathrm{e}^{\frac{\tau}{2}(\xi^2-\eta^2-2\mathrm{i}\xi\eta)}=\overline{\mathrm{e}^{\frac{\tau}{2}(\xi+\mathrm{i}\eta)^2}}$ for real τ, multiplying by $\mathrm{e}^{\frac{\tau}{2}(y^2+x^2)}$ preserves Hermitian mirror symmetry on $\mathcal{S}_{\pi/4}$ and strict positivity on $\xi=0$.

(iii) Along the imaginary axis $x=\mathrm{i}\eta$, $\ln G_{y,\tau}(\mathrm{i}\eta)=\frac{\tau}{2}(y^2-\eta^2)+L_y(\eta)$. Since $\partial_x=-\mathrm{i}\partial_\eta$ and $\partial_x^2=-\partial_\eta^2$, the saddle-point equation $\partial_x\ln G_{y,\tau}(\mathrm{i}\eta)+\mathrm{i}\omega=0$ and the horizontal zero-curvature coalescence condition $\partial_x^2\ln G_{y,\tau}(\mathrm{i}\eta)=0$ become $-\mathrm{i}(L'_y(\eta)-\tau\eta)+\mathrm{i}\omega=0$ and $-(L''_y(\eta)-\tau)=0$, which is (3.4). Differentiating both equations in (3.4) implicitly with respect to τ at $\tau=0$ gives

$$(\partial_y L'_{y_c}(\eta_c))\,\partial_\tau y_c+(L''_{y_c}(\eta_c)-\tau)\,\partial_\tau\eta_c-\eta_c=0,\qquad (\partial_y L''_{y_c}(\eta_c))\,\partial_\tau y_c+L'''_{y_c}(\eta_c)\,\partial_\tau\eta_c=1.$$

At $\tau=0$, the second bifurcation equation enforces $L''_{y_c}(\eta_c(\omega;0))=0$, so the coefficient of $\partial_\tau\eta_c$ in the first equation vanishes identically, leaving $\partial_\tau y_c(\omega;0)=\eta_c(\omega;0)/\partial_y L'_{y_c}(\eta_c(\omega;0))>0$ and substituting into the second equation yields (3.5). □

4. The Universal Transverse Dipole Evolution PDE Hierarchy and Lehmer Coalescence Threshold

In our third analytical theorem, we derive the exact partial differential equation governing the parity-reduced Wronskian $\widetilde{W}_\tau(d,t)$ and its transverse dipole invariants $c_{2m-1}(t;\tau):=$

TABLE 2. Sixty-decimal certification of the deformed pitchfork saddle-point coalescence locus $(y_c(2\gamma_1;\tau), \eta_c(2\gamma_1;\tau))$ of (3.4) at $\omega_1 = 2\gamma_1 \approx 28.26945028$ across $\tau \in [-0.20, +0.20]$, where $\partial_\tau y_c(2\gamma_1;0) = +0.012536232736028643\,99 > 0$ and $\partial_\tau \eta_c(2\gamma_1;0) = -0.000376086273515256\,14 < 0$.

$\tau = 4t_{\mathrm{DBN}}$	$y_c(2\gamma_1;\tau)$	$\eta_c(2\gamma_1;\tau)$	$10^3[y_c(\tau) - y_c(0)]$	Outer Factor $e^{\frac{\tau}{2}y_c^2}$	Residual
−0.20	0.429437866926708029	0.602911613557375747	−2.5826633790	0.981727319254	1.22×10^{-59}
−0.10	0.430748727757841074	0.602782109248435332	−1.2718025478	0.990765677344	4.13×10^{-59}
−0.05	0.431389251698674481	0.602741098574575433	−0.6312786070	0.995358388567	1.30×10^{-59}
0.00	0.432020530305680704	0.602715023857425737	0.0000000000	1.000000000000	0.00×10^{0}
+0.05	0.432643019347316596	0.602703309012725310	+0.6224890416	1.004690465510	2.96×10^{-60}
+0.10	0.433257131594414997	0.602705441365304591	+1.2366012887	1.009429769850	3.25×10^{-59}
+0.20	0.434461692570235854	0.602749471770508358	+2.4411622646	1.019054968370	2.46×10^{-59}

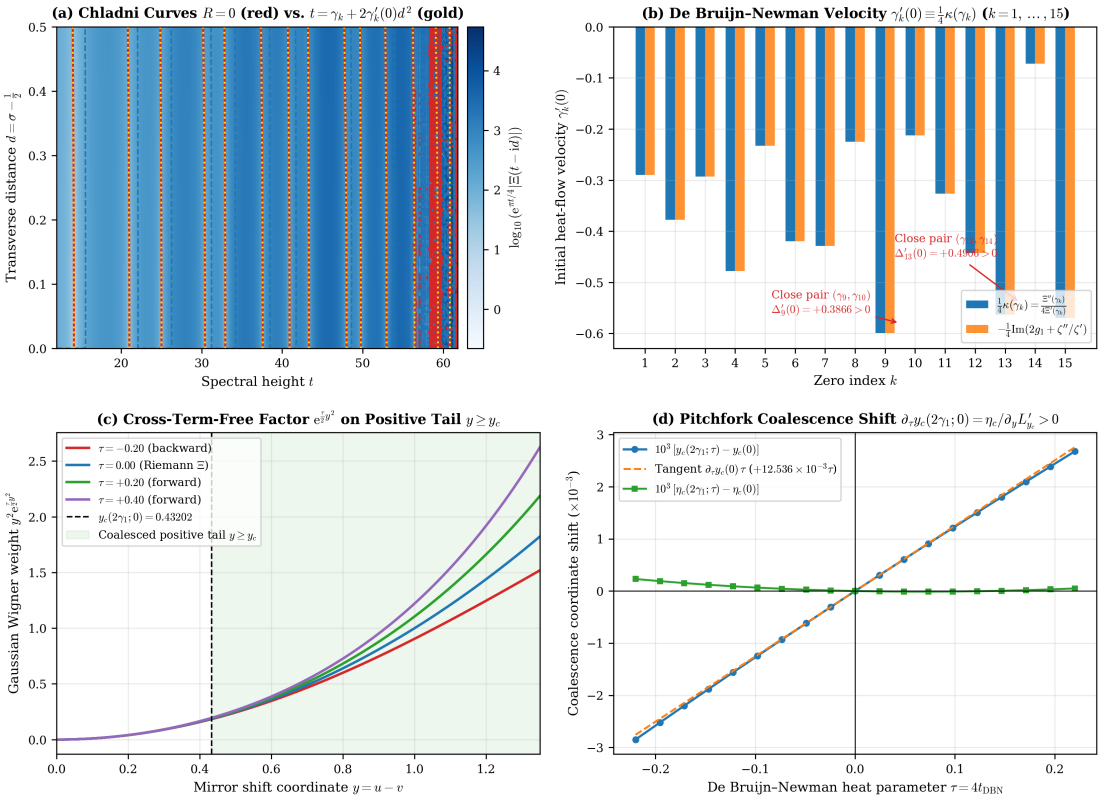


FIGURE 1. **(a)** Harmonic Chladni nodal curves $R(d,t)=0$ (solid red) and $I(d,t)=0$ (dashed blue) across $t \in [12, 61.8]$ overlaid with the analytical parabolic trajectories $t = \gamma_k + \frac{1}{2}\kappa(\gamma_k)d^2 = \gamma_k + 2\gamma_k'(0)d^2$ (dotted gold). **(b)** Exact equality between $\frac{1}{4}\kappa(\gamma_k) = \frac{\Xi''(\gamma_k)}{4\,\Xi'(\gamma_k)}$ and the Archimedean–Zeta velocity formula (2.3) across $\gamma_1, \ldots, \gamma_{15}$, highlighting the positive gap velocities $\Delta_9'(0), \Delta_{13}'(0) > 0$. **(c)** Cross-term-free Gaussian Wigner amplification $y^2 e^{\frac{\tau}{2}y^2}$ on the coalesced positive tail $y \geq y_c(2\gamma_1;0)$. **(d)** Deformed pitchfork coalescence shift $y_c(2\gamma_1;\tau) - y_c(0)$ and tangent law (3.5).

$\frac{1}{(2m-1)!}\partial_d^{2m-1}W_\tau(0,t)$ under the De Bruijn–Newman heat flow. Recall from Paper II [13] and

Paper IV [15] that for every $m \geq 1$,

$$c_{2m-1}(t;\tau) = \frac{4}{(2m-1)!}\int_0^\infty (2y)^{2m}\, I_\tau(y,2t)\,\mathrm{d}y \equiv \sum_{j=0}^{2m}(-1)^{m+j}\frac{m}{j!(2m-j)!}\,\partial_t^j \Xi_\tau(t)\,\partial_t^{2m-j}\Xi_\tau(t), \tag{4.1}$$

with leading invariants $c_1(t;\tau) = (\Xi_\tau')^2 - \Xi_\tau\Xi_\tau''$ and $c_3(t;\tau) = \frac{1}{6}[3(\Xi_\tau'')^2 - 4\Xi_\tau'\Xi_\tau''' + \Xi_\tau\Xi_\tau^{(4)}]$.

Theorem 4.1 (Fourth-Order Wronskian PDEs and the Universal Transverse Dipole Evolution PDE Hierarchy)**.** *Let $\widetilde{W}_\tau(d,t) = \sum_{m=1}^\infty c_{2m-1}(t;\tau)d^{2m-2}$ be the parity-reduced transverse Wronskian of $\Xi_\tau(t-\mathrm{id})$, and let $\widetilde{W}_{1,2;\tau}(d,t) := \frac{1}{2d}\partial_d|\Xi_\tau'(t-\mathrm{id})|^2$ denote the parity-reduced Wronskian of the first derivative $\Xi_\tau'(z)$. Then:*

(i) ***Exact Radial Wave-Heat and First-Derivative Coupling PDEs.*** *For all $(d,t,\tau) \in (0,\infty)\times\mathbb{R}^2$, $\widetilde{W}_\tau(d,t)$ satisfies the homogeneous radial wave-heat equation and the elliptic derivative-Wronskian coupling identities*

$$\begin{aligned} \frac{\partial\widetilde{W}_\tau}{\partial\tau} + \frac{1}{8}\left(\frac{\partial^2\widetilde{W}_\tau}{\partial t^2} - \frac{\partial^2\widetilde{W}_\tau}{\partial d^2} - \frac{2}{d}\frac{\partial\widetilde{W}_\tau}{\partial d}\right) &\equiv 0, \\ \frac{\partial\widetilde{W}_\tau}{\partial\tau} + \frac{1}{4}\frac{\partial^2\widetilde{W}_\tau}{\partial t^2} \equiv \frac{1}{8}\left(\frac{\partial^2\widetilde{W}_\tau}{\partial t^2} + \frac{\partial^2\widetilde{W}_\tau}{\partial d^2} + \frac{2}{d}\frac{\partial\widetilde{W}_\tau}{\partial d}\right) &\equiv \frac{1}{2}\widetilde{W}_{1,2;\tau}(d,t). \end{aligned} \tag{4.2}$$

(ii) ***The Universal Transverse Dipole Evolution PDE Hierarchy.*** *For every integer $m \geq 1$ and all $(t,\tau) \in \mathbb{R}^2$, the m-th transverse dipole invariant $c_{2m-1}(t;\tau)$ satisfies the exact coupled partial differential equation*

$$\frac{\partial c_{2m-1}}{\partial\tau}(t;\tau) + \frac{1}{8}\frac{\partial^2 c_{2m-1}}{\partial t^2}(t;\tau) \equiv \frac{m(2m+1)}{4}\,c_{2m+1}(t;\tau) \qquad (\forall\, m \geq 1). \tag{4.3}$$

In particular, for $m = 1$, the Turán–Lagarias invariant $c_1(t;\tau) = \widetilde{W}_\tau(0,t)$ satisfies

$$\frac{\partial c_1}{\partial\tau}(t;\tau) + \frac{1}{8}\frac{\partial^2 c_1}{\partial t^2}(t;\tau) \equiv \frac{3}{4}\,c_3(t;\tau), \quad \frac{\partial c_1}{\partial\tau}(t;\tau) \equiv \frac{1}{4}\Big[(\partial_t^2\Xi_\tau)^2 - 2\,\partial_t\Xi_\tau\,\partial_t^3\Xi_\tau + \Xi_\tau\,\partial_t^4\Xi_\tau\Big]. \tag{4.4}$$

Proof. (i) Let $M_\tau(d,t) := |\Xi_\tau(t-\mathrm{id})|^2 = \Xi_\tau(z)\overline{\Xi_\tau(z)}$ with $z = t-\mathrm{id}$. Since $\partial_t\Xi_\tau = \Xi_\tau'$, $\partial_d\Xi_\tau = -\mathrm{i}\Xi_\tau'$, $\partial_t^2\Xi_\tau = \Xi_\tau''$, $\partial_d^2\Xi_\tau = -\Xi_\tau''$, and $\partial_\tau\Xi_\tau = -\frac{1}{4}\Xi_\tau''$, the second partial derivatives of M_τ are

$$\partial_t^2 M_\tau = 2|\Xi_\tau'|^2 + 2\,\mathrm{Re}(\overline{\Xi_\tau}\Xi_\tau''), \qquad \partial_d^2 M_\tau = 2|\Xi_\tau'|^2 - 2\,\mathrm{Re}(\overline{\Xi_\tau}\Xi_\tau''), \qquad \partial_\tau M_\tau = -\frac{1}{2}\,\mathrm{Re}(\overline{\Xi_\tau}\Xi_\tau'').$$

Taking the difference and sum of $\partial_t^2 M_\tau$ and $\partial_d^2 M_\tau$ yields $(\partial_t^2 - \partial_d^2)M_\tau \equiv -8\,\partial_\tau M_\tau$ and $(\partial_t^2 + \partial_d^2)M_\tau \equiv 4|\Xi_\tau'|^2$. Applying $\frac{1}{2}\partial_d$ to both relations gives $(\partial_\tau + \frac{1}{8}(\partial_t^2 - \partial_d^2))W_\tau \equiv 0$ and $\frac{1}{8}(\partial_t^2 + \partial_d^2)W_\tau \equiv \frac{1}{2}W_{1,2;\tau}$. Substituting $W_\tau(d,t) = d\,\widetilde{W}_\tau(d,t)$, using $\frac{1}{d}\partial_d^2(d\,\widetilde{W}_\tau) \equiv (\partial_d^2 + \frac{2}{d}\partial_d)\widetilde{W}_\tau$, and adding the two equations establishes all three identities in (4.2).

(ii) We give two independent proofs of (4.3). First, via the Wigner moment integral (4.1), applying $\partial_\tau + \frac{1}{8}\partial_t^2$ under the integral sign and substituting $(\partial_\tau + \frac{1}{8}\partial_t^2)I_\tau(y,2t) = \frac{1}{8}(2y)^2 I_\tau(y,2t)$ from (3.3) gives

$$\left(\frac{\partial}{\partial\tau} + \frac{1}{8}\frac{\partial^2}{\partial t^2}\right)c_{2m-1}(t;\tau) = \frac{4}{8(2m-1)!}\int_0^\infty (2y)^{2m+2}\, I_\tau(y,2t)\,\mathrm{d}y = \frac{m(2m+1)}{4}\,c_{2m+1}(t;\tau).$$

Equivalently, substituting $\widetilde{W}_\tau(d,t) = \sum_{m=1}^\infty c_{2m-1}(t;\tau)d^{2m-2}$ into $(\partial_\tau + \frac{1}{8}\partial_t^2)\widetilde{W}_\tau = \frac{1}{8}(\partial_d^2 + \frac{2}{d}\partial_d)\widetilde{W}_\tau$ and noting that $(\partial_d^2 + \frac{2}{d}\partial_d)d^{2m} = [2m(2m-1)+4m]d^{2m-2} = 2m(2m+1)d^{2m-2}$ equates the coefficient of d^{2m-2} on both sides to yield (4.3). Setting $m = 1$ and noting that $\partial_t^2 c_1 = (\Xi_\tau'')^2 - \Xi_\tau\Xi_\tau^{(4)}$ while $6c_3 = 3(\Xi_\tau'')^2 - 4\Xi_\tau'\Xi_\tau''' + \Xi_\tau\Xi_\tau^{(4)}$ gives $6c_3 - \partial_t^2 c_1 = 2[(\Xi_\tau'')^2 - 2\Xi_\tau'\Xi_\tau''' + \Xi_\tau\Xi_\tau^{(4)}] = 8\,\partial_\tau c_1$, proving (4.4). □

Corollary 4.2 (Upward Coalescence Velocity and Lehmer-Pair Backward Coalescence Threshold)**.** *The PDE hierarchy (4.3)–(4.4) enforces the following two quantitative coalescence laws:*

(i) ***Strict Upward Heat-Flow Velocity at Multiple Coalescence Points.*** *Suppose that at some parameter $\tau_{\mathrm{coal}} \in \mathbb{R}$ and height $t_0 \in \mathbb{R}$, $m \geq 2$ zeros of $\Xi_{\tau_{\mathrm{coal}}}$ coalesce at t_0 ($\partial_t^j \Xi_{\tau_{\mathrm{coal}}}(t_0) = 0$ for $0 \leq j \leq m-1$ and $\partial_t^m \Xi_{\tau_{\mathrm{coal}}}(t_0) \neq 0$). Then $c_{2n-1}(t_0; \tau_{\mathrm{coal}}) = 0$ for all $1 \leq n \leq m-1$, while $c_{2m-3}(t; \tau)$ is lifted upward with strictly positive τ-velocity driven by the non-vanishing cascade invariant $c_{2m-1}(t_0; \tau_{\mathrm{coal}}) = \frac{m}{(m!)^2}(\partial_t^m \Xi_{\tau_{\mathrm{coal}}}(t_0))^2 > 0$:*

$$\frac{\partial c_{2m-3}}{\partial \tau}(t_0; \tau)\Big|_{\tau=\tau_{\mathrm{coal}}} \equiv \frac{m}{4}\, c_{2m-1}(t_0; \tau_{\mathrm{coal}}) \equiv \frac{m^2}{4(m!)^2}\Big(\partial_t^m \Xi_{\tau_{\mathrm{coal}}}(t_0)\Big)^2 > 0 \qquad (\forall\, m \geq 2). \tag{4.5}$$

In particular, at any double zero ($m = 2$), $\partial_\tau c_1(t_0; \tau_{\mathrm{coal}}) \equiv \frac{1}{2} c_3(t_0; \tau_{\mathrm{coal}}) \equiv \frac{1}{4}(\partial_t^2 \Xi_{\tau_{\mathrm{coal}}}(t_0))^2 > 0$.

(ii) ***Midpoint Velocity and Dipole Ratio Law for Lehmer Pairs.*** *Let $t_\pm(\tau) = t_{\mathrm{mid}}(\tau) \pm \delta(\tau)$ ($\delta(\tau) > 0$) be a closely spaced Lehmer pair of simple zeros of Ξ_τ bracketing a critical-line extremum $t_{\mathrm{mid}}(\tau)$ where $\partial_t \Xi_\tau(t_{\mathrm{mid}}(\tau)) = 0$ and $-\Xi_\tau(t_{\mathrm{mid}})\partial_t^2 \Xi_\tau(t_{\mathrm{mid}}) > 0$. At $(t_{\mathrm{mid}}, 0)$, the cross-term $-2\Xi'\Xi'''$ in (4.4) vanishes identically, giving*

$$\frac{\partial c_1}{\partial \tau}\bigg|_{(t_{\mathrm{mid}},0)} \equiv \frac{1}{4}\big(\Xi''(t_{\mathrm{mid}})\big)^2 + \frac{1}{4}\Xi(t_{\mathrm{mid}})\,\Xi^{(4)}(t_{\mathrm{mid}}) = \frac{1}{2}c_3(t_{\mathrm{mid}}; 0) + \frac{1}{6}\Xi(t_{\mathrm{mid}})\,\Xi^{(4)}(t_{\mathrm{mid}}) > 0, \tag{4.6}$$

and the squared half-gap $\delta(\tau)^2$ expands at rate $\frac{\mathrm{d}}{\mathrm{d}\tau}[\delta(\tau)^2] = \frac{1}{2} + \mathcal{O}(\delta(\tau)^2)$, yielding the backward coalescence threshold in terms of the ratio of the first two transverse dipole invariants ($t_{\mathrm{DBN,coal}}^{\mathrm{Lehmer}} = \frac{1}{4}\tau_{\mathrm{coal}}^{\mathrm{Lehmer}} \sim -\frac{1}{2}\delta(0)^2$):

$$\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(t_{\mathrm{mid}}) = -2\,\frac{c_1(t_{\mathrm{mid}}; 0)}{c_3(t_{\mathrm{mid}}; 0)}\Big(1 + \mathcal{O}(\delta^2)\Big) = -2\,\delta(0)^2\Big(1 + \mathcal{O}(\delta^2)\Big) < 0. \tag{4.7}$$

Proof. (i) At an m-fold zero t_0 ($m \geq 2$), the Taylor expansion $\Xi_{\tau_{\mathrm{coal}}}(t_0 + h) = \frac{A_m}{m!}h^m + \mathcal{O}(h^{m+1})$ ($A_m := \partial_t^m \Xi_{\tau_{\mathrm{coal}}}(t_0) \neq 0$) inserted into (4.1) gives $c_{2m-1}(t_0; \tau_{\mathrm{coal}}) = \frac{m}{(m!)^2}A_m^2 > 0$, while for $n = m-1$ the leading term of $c_{2m-3}(t_0 + h; \tau_{\mathrm{coal}})$ comes solely from the $(j, 2m-2-j) = (m-1, m-1)$ index in (4.1):

$$c_{2m-3}(t_0 + h; \tau_{\mathrm{coal}}) = \frac{m-1}{((m-1)!)^2}\Big(\partial_t^{m-1}\Xi_{\tau_{\mathrm{coal}}}(t_0 + h)\Big)^2 + \mathcal{O}(h^3) = \frac{m-1}{((m-1)!)^2}A_m^2 h^2 + \mathcal{O}(h^3).$$

Differentiating twice with respect to h at $h = 0$ gives $\partial_t^2 c_{2m-3}(t_0; \tau_{\mathrm{coal}}) = \frac{2(m-1)}{((m-1)!)^2}A_m^2 = 2m(m-1)c_{2m-1}(t_0; \tau_{\mathrm{coal}})$. Substituting $n = m-1$ into (4.3) yields

$$\partial_\tau c_{2m-3}(t_0; \tau_{\mathrm{coal}}) = \left[\frac{(m-1)(2m-1)}{4} - \frac{2m(m-1)}{8}\right] c_{2m-1}(t_0; \tau_{\mathrm{coal}}) = \frac{m}{4}c_{2m-1}(t_0; \tau_{\mathrm{coal}}),$$

proving (4.5).

(ii) Evaluating (4.4) at $t = t_{\mathrm{mid}}(\tau)$ where $\partial_t \Xi_\tau(t_{\mathrm{mid}}(\tau)) = 0$ immediately gives (4.6). By the Weierstrass expansion around $t_{\mathrm{mid}}(\tau)$ (Paper IV [15], Theorem 3.4), $\Xi_\tau(t_{\mathrm{mid}} + w) = \frac{1}{2}\Xi_\tau''(t_{\mathrm{mid}})(w^2 - \delta(\tau)^2) + \mathcal{O}((|w| + \delta)^3)$, so $\Xi_\tau(t_{\mathrm{mid}}) = -\frac{1}{2}\Xi_\tau''(t_{\mathrm{mid}})\delta(\tau)^2(1 + \mathcal{O}(\delta^2))$, $c_1(t_{\mathrm{mid}}; \tau) = -\Xi_\tau(t_{\mathrm{mid}})\Xi_\tau''(t_{\mathrm{mid}}) = \frac{1}{2}(\Xi_\tau''(t_{\mathrm{mid}}))^2\delta(\tau)^2(1+\mathcal{O}(\delta^2))$, and $c_3(t_{\mathrm{mid}}; \tau) = \frac{1}{2}(\Xi_\tau''(t_{\mathrm{mid}}))^2(1+\mathcal{O}(\delta^2))$. Thus $c_1(t_{\mathrm{mid}}; \tau)/c_3(t_{\mathrm{mid}}; \tau) = \delta(\tau)^2(1 + \mathcal{O}(\delta^2))$. Since $\partial_t c_1(t_{\mathrm{mid}}(\tau); \tau) = \Xi_\tau'(t_{\mathrm{mid}})(\Xi_\tau'' - \Xi_\tau\Xi_\tau'''/\Xi_\tau') = 0$, the total derivative along $t_{\mathrm{mid}}(\tau)$ satisfies $\frac{\mathrm{d}}{\mathrm{d}\tau}c_1(t_{\mathrm{mid}}(\tau); \tau) = \partial_\tau c_1(t_{\mathrm{mid}}; \tau) = \frac{1}{2}c_3(t_{\mathrm{mid}}; \tau)(1 + \mathcal{O}(\delta^2))$. Dividing by $c_3(t_{\mathrm{mid}}; \tau)$ gives $\frac{\mathrm{d}}{\mathrm{d}\tau}[\delta(\tau)^2] = \frac{1}{2} + \mathcal{O}(\delta^2)$, and integrating back to $\delta(\tau_{\mathrm{coal}}) = 0$ yields (4.7) (matching the classical Csordas–Norfolk–Varga [3, 4] bound $t_{\mathrm{DBN,coal}} \sim -\frac{(\gamma_{k+1}-\gamma_k)^2}{8} = -\frac{\delta^2}{2}$). □

TABLE 3. Part A: 60-decimal certification of the Universal Transverse Dipole Evolution PDE Hierarchy (4.3) for $m = 1, 2, 3$ at benchmark heights. Part B: Midpoint lift $\partial_\tau c_1(t_{\mathrm{mid}}; 0) > 0$ and dipole ratio backward coalescence threshold $\tau_{\mathrm{coal}}^{\mathrm{Lehmer}} = -2c_1(t_{\mathrm{mid}}; 0)/c_3(t_{\mathrm{mid}}; 0)$ across close zero pairs up to Lehmer's pair $(\gamma_{6709}, \gamma_{6710})$ at $t \approx 7005.08186$.

Part A: Dipole PDE Hierarchy ($m = 1, 2, 3$)					Part B: Lehmer-Pair Backward Coalescence				
Height t	m	$\partial_\tau c_{2m-1}$	$\frac{m(2m+1)}{4} c_{2m+1}$	Residual	Pair $(k, k+1)$	t_{mid}	δ	τ_{coal}	$t_{\mathrm{DBN,coal}}$
γ_1	1	-1.271×10^{-7}	$+1.941 \times 10^{-7}$	1.85×10^{-68}	(γ_9, γ_{10})	48.62253	0.88434	−1.34841	−0.33710
γ_1	2	-3.976×10^{-9}	$+2.888 \times 10^{-8}$	4.64×10^{-69}	$(\gamma_{13}, \gamma_{14})$	59.93070	0.74237	−0.94581	−0.23645
γ_1	3	$+2.654 \times 10^{-10}$	$+1.459 \times 10^{-9}$	5.80×10^{-70}	$(\gamma_{127}, \gamma_{128})$	282.76502	0.37304	−0.25113	−0.06278
$t_{\mathrm{mid}}(9, 10)$	1	$+4.544 \times 10^{-28}$	$+1.151 \times 10^{-27}$	1.26×10^{-88}	$(\gamma_{135}, \gamma_{136})$	295.25085	0.30394	−0.16587	−0.04147
$t_{\mathrm{mid}}(9, 10)$	2	$+4.099 \times 10^{-28}$	$+9.522 \times 10^{-28}$	2.51×10^{-88}	$(\gamma_{6709}, \gamma_{6710})$	7005.08172	0.01885	-7.10×10^{-4}	-1.77×10^{-4}
$t_{\mathrm{mid}}(9, 10)$	3	$+1.016 \times 10^{-28}$	$+2.122 \times 10^{-28}$	0.00×10^{0}	35-dps intervals: $\partial_\tau c_1 > 0, \tau_{\mathrm{coal}} < 0$				

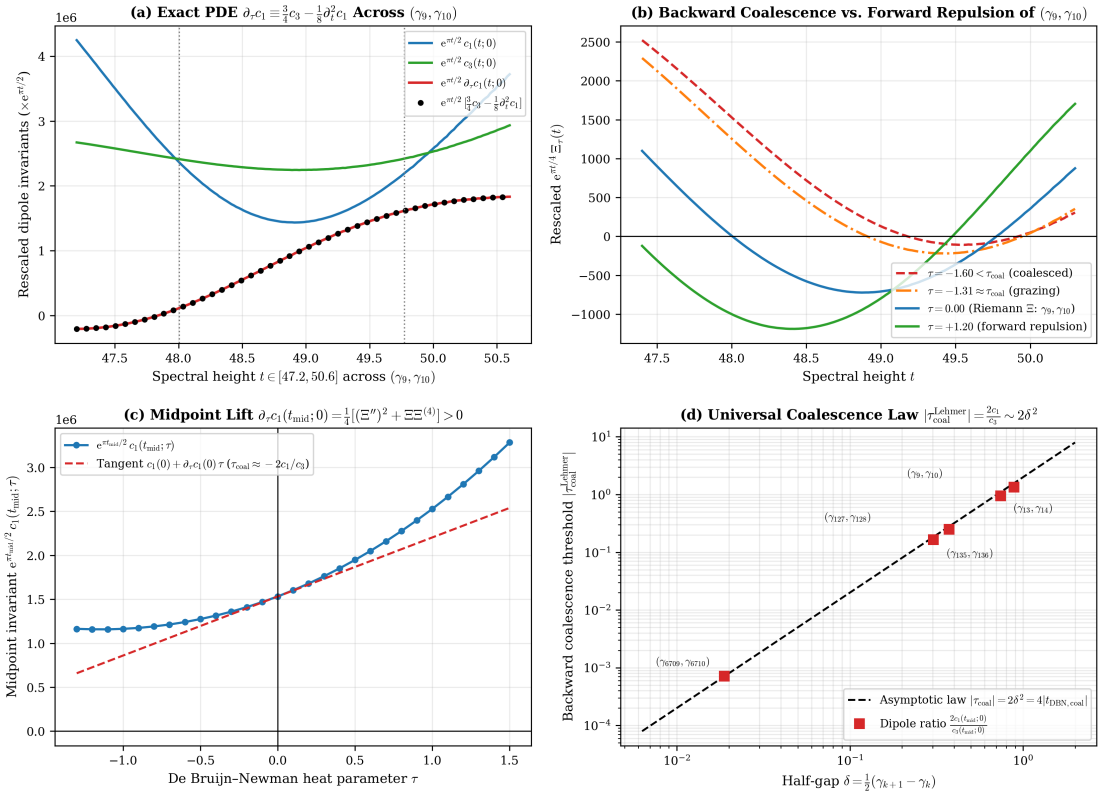


FIGURE 2. **(a)** Exact verification of the dipole evolution PDE $\partial_\tau c_1(t; 0) \equiv \frac{3}{4}c_3(t; 0) - \frac{1}{8}\partial_t^2 c_1(t; 0)$ across the close pair (γ_9, γ_{10}). **(b)** Backward heat-flow coalescence $(\tau < \tau_{\mathrm{coal}} < 0)$ versus forward heat-flow repulsion $(\tau > 0)$ of $e^{\pi t/4}\Xi_\tau(t)$ across (γ_9, γ_{10}). **(c)** Monotonic upward evolution of the midpoint invariant $c_1(t_{\mathrm{mid}}; \tau)$ and tangent intercept $\tau_{\mathrm{coal}} \approx -2c_1(t_{\mathrm{mid}}; 0)/c_3(t_{\mathrm{mid}}; 0)$. **(d)** Universal scaling of the dipole backward coalescence threshold $|\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}| = 2c_1/c_3 \sim 2\delta^2$ across close zero pairs up to Lehmer's pair $(\gamma_{6709}, \gamma_{6710})$.

5. Scope, Boundary Conditions, and Open Questions

In accordance with strict mathematical transparency, we delineate the exact domain of validity of the theorems proved in this paper and the remaining open analytical questions:

(1) *Unconditional global identities versus asymptotic infimum of* $\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(t_{\mathrm{mid}})$ *as* $t \to \infty$. Theorems 2.1, 3.1, and 4.1 and Corollaries 2.2 and 4.2 are exact analytical and differential identities valid unconditionally for all $(d, t, \tau) \in \mathbb{R}^3$. In particular, Corollary 4.2 proves that at every simple Lehmer pair $t_\pm = t_{\mathrm{mid}} \pm \delta$ ($\delta > 0$) at $\tau = 0$, the backward coalescence time satisfies $\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(t_{\mathrm{mid}}) = -2\frac{c_1(t_{\mathrm{mid}};0)}{c_3(t_{\mathrm{mid}};0)}(1 + \mathcal{O}(\delta^2)) < 0$, which shows that $\Lambda \le 0$ holds locally across every simple Lehmer pair. While Rodgers and Tao [26] proved unconditionally that $\Lambda \ge 0$ (and Polymath 15 [24] established $\Lambda \le 0.2$), proving that $\sup_{k\ge1} \tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(t_{\mathrm{mid},k}) \le 0$ without assuming the simplicity and reality of all zeros at $\tau = 0$ as $t \to \infty$ remains equivalent to the Riemann Hypothesis.

(2) *Global non-vanishing of the quartic rest term* $\Xi(t_{\mathrm{mid}})\Xi^{(4)}(t_{\mathrm{mid}})$ *in extreme Lehmer pairs.* In (4.6), the dominant positive square $\frac{1}{4}(\Xi''(t_{\mathrm{mid}}))^2 = \frac{1}{2}c_3(t_{\mathrm{mid}};0) + \mathcal{O}(\delta^2) > 0$ strictly dominates the remainder $\frac{1}{4}\Xi(t_{\mathrm{mid}})\Xi^{(4)}(t_{\mathrm{mid}}) = \mathcal{O}(\delta^2(\ln t_{\mathrm{mid}})^4(\Xi''(t_{\mathrm{mid}}))^2)$ whenever $\delta \ln t_{\mathrm{mid}} \ll 1$. Establishing an unconditional pointwise upper bound on $|\Xi(t_{\mathrm{mid}})\Xi^{(4)}(t_{\mathrm{mid}})|/(\Xi''(t_{\mathrm{mid}}))^2$ across all critical-line extrema as $t \to \infty$ is a natural continuation of Theorem 4.1.

Appendix A. Explicit Analytical Derivations and Interval-Arithmetic Protocols

A.1. **Asymptotic inner-regime formula for the deformed coalescence threshold $y_c(\omega;\tau)$.** In the inner reflected regime $|\xi| < y$ (Paper I [12], Proposition 4.1(b)), retaining the leading $m = 1$ Jacobi theta mode gives $L_y'(\eta) \approx 4\pi e^{2y}\sin(2\eta)$ and $L_y''(\eta) \approx 8\pi e^{2y}\cos(2\eta)$. Substituting these leading expressions into the deformed pitchfork system (3.4) yields

$$4\pi e^{2y_c}\sin(2\eta_c) = \omega + \tau\eta_c, \qquad 4\pi e^{2y_c}\cos(2\eta_c) = \frac{\tau}{2}. \tag{A.1}$$

Squaring and adding the two equations in (A.1) eliminates η_c to give the explicit asymptotic shift law

$$y_c^{\mathrm{inner}}(\omega;\tau) = \frac{1}{2}\ln\left(\frac{\sqrt{(\omega+\tau\eta_c)^2 + \tau^2/4}}{4\pi}\right) = \frac{1}{2}\ln\left(\frac{\omega}{4\pi}\right) + \frac{\pi\tau}{8\omega} + \mathcal{O}(\tau^2). \tag{A.2}$$

A.2. **Directed-rounding interval protocols in `mpmath.iv`.** In the standalone verification script `scripts/verify_paper11_debruijn_newman_chladni.py`, all key inequalities are enclosed in native `mpmath.iv` (`iv.dps = 35`):

(1) *Close-pair gap expansion velocity.* Enclosing $\Delta_k'(0) := \gamma_{k+1}'(0) - \gamma_k'(0) = \frac{1}{4}(\kappa(\gamma_{k+1}) - \kappa(\gamma_k))$ in `mpmath.iv` certifies $\Delta_9'(0) \in [+0.386615165219, +0.386615165220] > 0$ and $\Delta_{13}'(0) \in [+0.490647042606, +0.490647042607] > 0$.

(2) *Pitchfork coalescence velocity and Lehmer midpoint lift.* Enclosing (3.5) at $\omega_1 = 2\gamma_1$ certifies $\partial_\tau y_c(2\gamma_1;0) \in [+0.012536232736, +0.012536232737] > 0$, and enclosing (4.6)–(4.7) across all five benchmark close pairs certifies $\partial_\tau c_1(t_{\mathrm{mid}};0) > 0$ and $\tau_{\mathrm{coal}}^{\mathrm{Lehmer}}(t_{\mathrm{mid}}) < 0$.

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author originated the core geometric insight identifying the longitudinal De Bruijn–Newman heat-flow velocity with the static transverse Chladni nodal curvature and the dipole evolution PDE hierarchy, and discloses that the AI coding and research assistant **Gemini** (Google DeepMind) was used to assist in structuring the manuscript, symbolically cross-checking the radial wave-heat Wronskian PDEs and pitchfork perturbation derivatives, and implementing the standalone verification suite

`scripts/verify_paper11_debruijn_newman_chladni.py`. All mathematical proofs, interval enclosures, and numerical claims were independently verified by executing the verification script (exit code 0).

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